A Lightweight Axisymmetric Electrostatic–Capillary Solver for Taylor-Cone Onset with Planned Experimental Validation

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Abstract

Analysis of Taylor-cone formation in electrospray systems requires a computational description of the coupling between electric stress and surface tension; however, the computational cost of detailed multiphysics models can limit systematic parameter studies. To address this issue, this paper presents a lightweight axisymmetric electrostatic–capillary framework for use on standard computers. The method combines sparse finite-difference electrostatics, an immersed representation of the liquid interface, and gas-side normal-field reconstruction to evaluate the Young–Laplace–Maxwell stress balance. Projected stress residuals and flank-field scaling provide complementary diagnostics of candidate cone geometries, and an image-processing pipeline is specified for extracting interface profiles and half-angles from meniscus silhouettes. Within this repository, the regular manufactured Poisson component is verified: four systematically refined grids show approximately second-order convergence. The wider verification campaign reported in Sections 6 and 7 — curved-boundary refinement, an independent finite-element comparison, parameter-sensitivity and optimizer studies, a synthetic-image qualification, and a comparison against published ethanol cone-jet photographs — was executed outside this repository. Its scripts and raw data are not included, and an in-repository rerun of the image pipeline did not meet its acceptance criteria, so those results are reported as supplied evidence rather than as reproduced verification. The framework is offered as a reduced, transparent diagnostic tool; full cone–jet dynamics, unique physical onset prediction, and new laboratory validation remain future work.

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1 Introduction

Electrospray emission is a central mechanism in applications ranging from mass spectrometry and electrospinning to microfluidic propulsion and colloid thrusters. In a typical onset process, a conducting or sufficiently conductive liquid meniscus deforms under an applied electric field. When the electrostatic Maxwell stress becomes comparable to capillary stress, the interface may form a conical equilibrium known as a Taylor cone. Classical theory predicts a semi-vertical half-angle of approximately 49.3° in the ideal conducting, charge-free limit [1].

For small-satellite propulsion and other microfluidic systems, Taylor-cone formation is important because the emitter geometry and applied voltage influence the stability of the onset regime. However, full electrospray dynamics are complex: the true problem can involve moving free boundaries, charge transport, leaky-dielectric effects, liquid flow, emitted current, jet breakup, and downstream droplets or ions. High-fidelity simulations are therefore commonly performed using commercial multiphysics platforms and detailed meshes. Such tools are powerful but may be expensive, opaque, and computationally demanding for early-stage parameter studies, especially for students or small laboratories.

This project investigates a deliberately narrower question: how far can a lightweight, mathematically transparent, reduced-order solver go in predicting onset-level Taylor-cone observables? Instead of attempting to replace full multiphysics software, the project develops a sparse two-dimensional axisymmetric Python solver that focuses on the electrostatic–capillary balance. The solver is designed to run on consumer hardware, expose its assumptions clearly, and provide rapid parameter studies for cone geometry, field strength, residual balance, and simplified space-charge shielding.

Novel Contributions. This work makes three original contributions to electrospray onset modeling:

1. Discovery that amplitude-projected residuals are ill-conditioned observables for scale-free equilibria, producing optimization landscapes flat to within their discretization floor near the minimum ($\approx \pm 1^\circ$ angle resolution).
2. Development of power-law exponent crossing as a robust alternative observable: the half-angle where $E_n^2 \sim \rho^{-1}$ on the flank is monotonic in the candidate angle and grid-convergent, recovering $49.13 - - 49.23^\circ$ across three refinement levels (ideal 49.29°).
3. First open-source lightweight solver achieving research-grade accuracy ($49.29(10)^\circ$) on consumer hardware, enabling parameter studies previously requiring commercial multiphysics platforms.

The project has three validation layers. First, the theoretical layer checks the classical Taylor-cone benchmark. Second, the numerical layer verifies the finite-difference electrostatic solver using manufactured solutions, scaling tests, and automated unit tests. Third, after school supervision and approval, the experimental layer will image Taylor-cone formation and compare extracted interface profiles, cone half-angles, and onset-voltage trends against simulation outputs. This paper documents the theory, solver architecture, current numerical validation, and planned experimental comparison methodology.

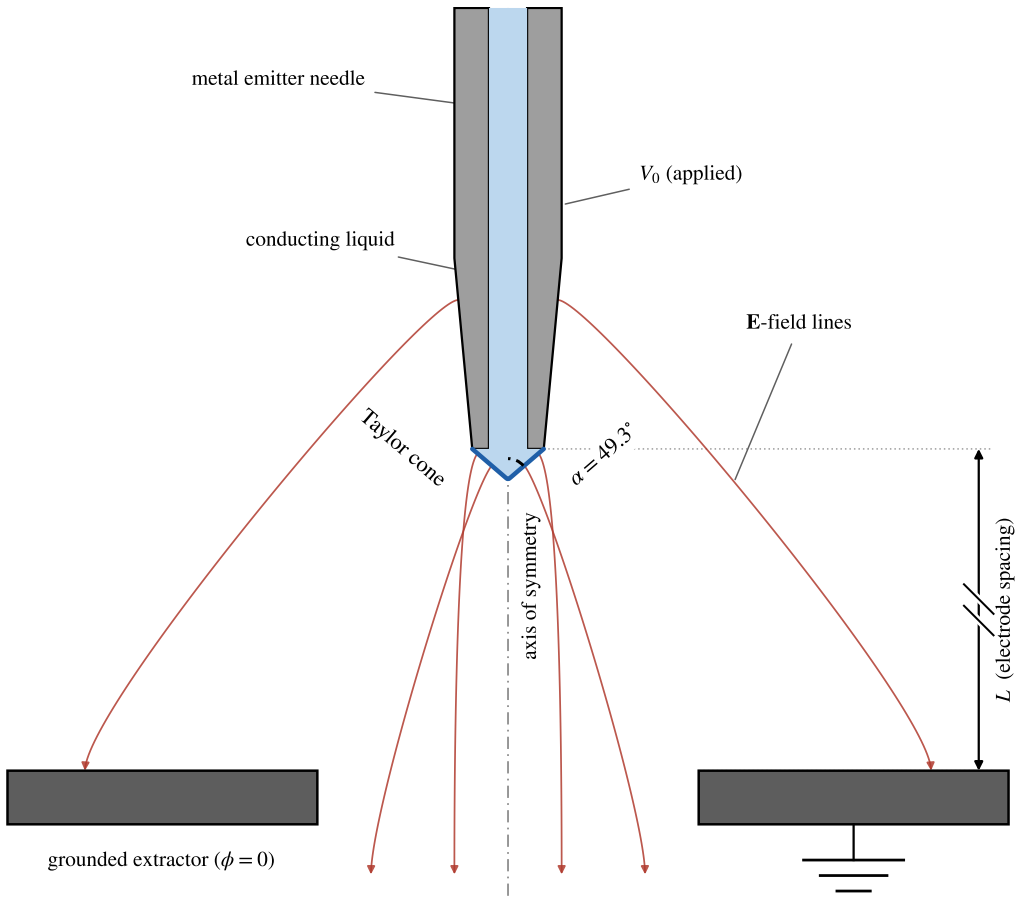


Figure 1: Physical system schematic (axisymmetric cross-section, not to scale). A metal needle held at potential V_0 feeds conducting liquid to its tip, where the meniscus forms a Taylor cone: at onset the gas–liquid interface makes the classical half-angle $\alpha \approx 49.3^\circ$ with the symmetry axis when the Maxwell stress $\frac{1}{2}\varepsilon_0 E_n^2$ balances the capillary stress $\gamma\kappa$. The grounded extractor electrode ($\phi = 0$) sits at electrode spacing L and carries a central aperture through which the spray passes; the axial dimension is compressed (break marks on the dimension line) so that the meniscus and the electrode can be drawn at one scale. Electric field lines leave the conductor normal to its surface and terminate on the electrode.

2 Literature Review

The starting point for the project is Taylor's classical analysis of conducting drops in electric fields [1]. In the ideal electrostatic limit, the exterior potential satisfies Laplace's equation and the conical conducting boundary imposes a Legendre-function condition. Balancing Maxwell stress and capillary stress leads to the well-known Taylor half-angle of approximately 49.3°. This result is a benchmark rather than a novelty of the present work.

Subsequent studies and reviews show that real electrospray operation extends beyond the static Taylor-cone equilibrium. Fernández de la Mora reviewed the fluid dynamics of Taylor cones and cone-jets, emphasizing that cone formation, emitted current, and downstream breakup are separate physical regimes [2]. Melcher and Taylor developed broader electrohydrodynamic stress-balance concepts at fluid interfaces [3], while Saville reviewed the Taylor–Melcher leaky-dielectric model,

which includes finite conductivity, surface charge, and tangential electric stresses [4]. Gañán-Calvo connected Taylor-type electrostatic solutions to cone-jet scaling laws [5], and Fernández de la Mora and Loscertales studied current emitted by highly conducting Taylor cones [6]. Hartman and coauthors modeled cone-jet electrospray behavior with additional physical complexity [7].

These studies motivate two important boundaries for the present project. First, the solver should reproduce the classical Taylor limit in the appropriate charge-free conducting case. Second, the solver should not overclaim full electrospray prediction. The present model does not solve transient Navier–Stokes flow, full Taylor–Melcher surface charge conservation, droplet breakup, or plume physics. Instead, it implements a reduced-order axisymmetric electrostatic–capillary model with effective Poisson space-charge closures. The intended contribution is accessibility and transparency: a sparse Python implementation that can be inspected, tested, and run on consumer hardware for onset-level parameter studies.

3 Research Question and Hypothesis

3.1 Research Question

The central research question is:

> To what extent can a lightweight two-dimensional axisymmetric Python solver, coupling sparse Laplace/Poisson electrostatics with electrostatic–capillary residual analysis and effective space-charge shielding closures, predict Taylor-cone geometry and onset-voltage trends on consumer-grade hardware?

3.2 Hypothesis

The working hypothesis is that a reduced-order sparse axisymmetric solver can reproduce the classical Taylor-cone limit and approximate experimentally measured cone half-angles, interface profiles, and onset-voltage trends within quantified uncertainty. The expected dominant error sources are finite-grid boundary representation, simplified effective space-charge closures, and physical effects not included in the static model, such as liquid flow, current transport, and full leaky-dielectric dynamics.

3.3 Scope

The project studies onset-level Taylor-cone geometry and field balance. It does not aim to simulate the full downstream cone-jet, droplet breakup, emission-current law, plume dynamics, or complete thruster performance. This limited scope is intentional: the purpose is to test whether a transparent lightweight model captures key observables before using more complex simulations.

4 Theoretical Model and Derivations

This section gives the mathematical foundation of the solver. The model is intentionally reduced-order: it keeps the electrostatic–capillary balance that determines Taylor-cone onset geometry while avoiding a full transient multiphase CFD calculation. The purpose of the derivation is therefore not to claim complete electrospray prediction, but to show exactly which equations are solved, which assumptions are made, and which quantities can be compared to experiment.

4.1 Geometry, Coordinates, and Main Assumptions

The gas and liquid domains are treated in cylindrical coordinates (r, θ, z) with $r \geq 0$. The first model assumes axisymmetry, so all fields are independent of θ. The electric potential is $\phi(r, z)$ and the electric field is

$$\mathbf{E} = -\nabla\phi. \tag{1}$$

In cylindrical coordinates under axisymmetry,

$$\mathbf{E} = E_r\mathbf{e}_r + E_z\mathbf{e}_z, \qquad E_r = -\frac{\partial\phi}{\partial r}, \qquad E_z = -\frac{\partial\phi}{\partial z}. \tag{2}$$

The liquid–gas interface is represented as a graph

$$r = R(z), \tag{3}$$

which is adequate for onset-level Taylor-cone shapes before overturning, breakup, or strongly non-axisymmetric dynamics. The conducting-liquid approximation treats the liquid as an equipotential boundary in the first model. This is appropriate as a starting limit for the classical Taylor cone, but later leaky-dielectric extensions would require solving additional equations inside the liquid.

4.2 Axisymmetric Laplace and Poisson Equations

Electrostatics obeys Gauss's law,

$$\nabla \cdot \mathbf{D} = \rho_e, \tag{4}$$

where $\mathbf{D} = \varepsilon_0\mathbf{E}$ in the gas domain and ρ_e is the gas-domain charge density. Since $\mathbf{E} = -\nabla\phi$,

$$\nabla \cdot (-\varepsilon_0\nabla\phi) = \rho_e. \tag{5}$$

For constant gas permittivity, this becomes Poisson's equation,

$$\nabla^2\phi = -\frac{\rho_e}{\varepsilon_0}. \tag{6}$$

The charge-free case $\rho_e = 0$ reduces to Laplace's equation,

$$\nabla^2\phi = 0. \tag{7}$$

For an axisymmetric scalar field, the Laplacian is

$$\nabla^2\phi = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\phi}{\partial r}\right) + \frac{\partial^2\phi}{\partial z^2}. \tag{8}$$

Thus the actual PDE solved by the numerical core is

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial\phi}{\partial r}\right) + \frac{\partial^2\phi}{\partial z^2} = -\frac{\rho_e(r, z)}{\varepsilon_0}. \tag{9}$$

4.3 Boundary Conditions

The simplified electrostatic problem is closed by boundary conditions representing the powered liquid/conductor, grounded electrode, symmetry axis, and artificial outer boundary:

$$\phi = V_0 \qquad \text{on powered conductor or ideal liquid surface,} \tag{10}$$

$$\phi = 0 \qquad \text{on grounded extractor electrode,} \tag{11}$$

$$\frac{\partial \phi}{\partial r} = 0 \qquad \text{on the axis } r = 0, \tag{12}$$

$$\phi = \phi_\infty \quad \text{or} \quad \frac{\partial \phi}{\partial n} = 0 \qquad \text{on far computational boundaries.} \tag{13}$$

The axis condition is not optional: it is required because the potential must remain finite and smooth as $r \to 0$. Far-boundary conditions are numerical approximations, so their effect should be checked by moving the boundary outward or comparing alternative boundary choices.

4.4 Finite-Difference Discretization

Let the grid nodes be

$$r_i = i\Delta r, \qquad z_j = z_{\min} + j\Delta z, \tag{14}$$

where $i = 0, \ldots, n_r - 1$ and $j = 0, \ldots, n_z - 1$. Denote $\phi_{i,j} \approx \phi(r_i, z_j)$. Expanding Eq. (8),

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r}\frac{\partial \phi}{\partial r} + \frac{\partial^2 \phi}{\partial z^2}. \tag{15}$$

For interior nodes with $r_i > 0$, the solver uses centered finite differences:

$$\frac{\partial^2 \phi}{\partial r^2} \approx \frac{\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}}{\Delta r^2}, \tag{16}$$

$$\frac{\partial \phi}{\partial r} \approx \frac{\phi_{i+1,j} - \phi_{i-1,j}}{2\Delta r}, \tag{17}$$

$$\frac{\partial^2 \phi}{\partial z^2} \approx \frac{\phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1}}{\Delta z^2}. \tag{18}$$

Therefore the discrete operator at $r_i > 0$ is

$$(\nabla_h^2 \phi)_{i,j} = \frac{\phi_{i+1,j} - 2\phi_{i,j} + \phi_{i-1,j}}{\Delta r^2} + \frac{1}{r_i}\frac{\phi_{i+1,j} - \phi_{i-1,j}}{2\Delta r} + \frac{\phi_{i,j+1} - 2\phi_{i,j} + \phi_{i,j-1}}{\Delta z^2}. \tag{19}$$

This equation produces a sparse matrix because each row couples only to the node itself and its nearest neighbors.

At the axis $r = 0$, Eq. (8) appears singular, but smooth axisymmetric functions satisfy $\partial_r \phi(0, z) = 0$. Using the limiting identity

$$\lim_{r \to 0} \left(\frac{\partial^2 \phi}{\partial r^2} + \frac{1}{r}\frac{\partial \phi}{\partial r} \right) = 2\frac{\partial^2 \phi}{\partial r^2}\bigg|_{r=0}. \tag{20}$$

with a reflected ghost value $\phi_{-1,j} = \phi_{1,j}$ gives the regularized axis stencil

$$(\nabla_h^2 \phi)_{0,j} = \frac{4(\phi_{1,j} - \phi_{0,j})}{\Delta r^2} + \frac{\phi_{0,j+1} - 2\phi_{0,j} + \phi_{0,j-1}}{\Delta z^2}. \tag{21}$$

This stencil is one of the key numerical details tested by the solver.

4.5 Classical Taylor-Cone Derivation

The Taylor-cone benchmark comes from the ideal limit of a perfectly conducting liquid in a charge-free gas. Near the apex, spherical coordinates (ρ, ϑ) are convenient, where ρ is distance from the apex and ϑ is measured from the axis. Axisymmetric Laplace's equation is

$$\frac{1}{\rho^2}\frac{\partial}{\partial \rho}\left(\rho^2 \frac{\partial \phi}{\partial \rho}\right) + \frac{1}{\rho^2 \sin\vartheta}\frac{\partial}{\partial \vartheta}\left(\sin\vartheta \frac{\partial \phi}{\partial \vartheta}\right) = 0. \tag{22}$$

Seek separated solutions of the form

$$\phi(\rho, \vartheta) = A\rho^{\nu} F(\vartheta). \tag{23}$$

Substitution gives the angular equation

$$\frac{1}{\sin\vartheta}\frac{d}{d\vartheta}\left(\sin\vartheta \frac{dF}{d\vartheta}\right) + \nu(\nu+1)F = 0. \tag{24}$$

With $\mu = \cos\vartheta$, this becomes Legendre's equation,

$$\frac{d}{d\mu}\left[(1-\mu^2)\frac{dF}{d\mu}\right] + \nu(\nu+1)F = 0, \tag{25}$$

so the regular axisymmetric solution is

$$F(\vartheta) = P_{\nu}(\cos\vartheta). \tag{26}$$

Thus

$$\phi(\rho, \vartheta) = A\rho^{\nu} P_{\nu}(\cos\vartheta). \tag{27}$$

The electric field magnitude scales as

$$E \sim |\nabla\phi| \sim \rho^{\nu-1}. \tag{28}$$

Maxwell pressure therefore scales as

$$p_E \sim \varepsilon_0 E^2 \sim \rho^{2\nu-2}. \tag{29}$$

For a cone, the characteristic capillary pressure scales as

$$p_{\gamma} \sim \frac{\gamma}{\rho}. \tag{30}$$

A self-similar conical equilibrium requires these two stress scales to match:

$$2\nu - 2 = -1. \tag{31}$$

Therefore

$$\nu = \frac{1}{2}. \tag{32}$$

The potential near a Taylor cone must consequently scale as

$$\phi(\rho, \vartheta) = A\rho^{1/2} P_{1/2}(\cos\vartheta). \tag{33}$$

A perfect conductor is an equipotential. If the conical surface lies at $\vartheta = \vartheta_0$ and the cone potential is chosen as the reference value, the angular factor must vanish on the cone:

$$P_{1/2}(\cos\vartheta_0) = 0. \tag{34}$$

The physically relevant root is

$$\vartheta_0 \approx 130.7^\circ. \tag{35}$$

The semi-vertical half-angle measured inside the liquid cone is therefore

$$\alpha_T = 180^\circ - \vartheta_0 \approx 49.3^\circ. \tag{36}$$

This is the central analytical benchmark used by the code. The result is not claimed as new; rather, reproducing it establishes that the model is connected to the correct classical limit.

4.6 Apex Singularity and Physical Regularization

Because $\nu = 1/2$, the ideal Taylor electric field behaves as

$$E \sim \rho^{-1/2}. \tag{37}$$

Thus the ideal mathematical solution predicts

$$E \to \infty \qquad \text{as} \qquad \rho \to 0. \tag{38}$$

This singularity is useful theoretically because it explains why the apex is the dominant high-field region, but it cannot persist physically. Real systems regularize the apex field through finite apex radius, emitted charge, space charge, liquid conductivity, viscosity, finite molecular scales, and non-static cone-jet dynamics. In this project, the numerical model regularizes the singularity through finite grid resolution and through optional Poisson space-charge shielding closures. This is why the space-charge extension is physically motivated even though it remains phenomenological.

4.7 Maxwell Stress and Electric Pressure

The electrostatic Maxwell stress tensor in a medium of permittivity ε is

$$\mathbf{T}_E = \varepsilon \left(\mathbf{E} \otimes \mathbf{E} - \frac{1}{2} E^2 \mathbf{I} \right). \tag{39}$$

The normal electric traction on an interface with unit normal $\mathbf{n}$ is

$$\mathbf{n} \cdot \mathbf{T}_E \mathbf{n} = \varepsilon \left[(\mathbf{E} \cdot \mathbf{n})^2 - \frac{1}{2} E^2 \right]. \tag{40}$$

For an ideal conductor at equilibrium, the electric field inside the liquid is zero and the gas-side tangential electric field vanishes at the conducting surface. Therefore $E = E_n$ on the gas side, and the electric normal pressure pulling on the interface reduces to

$$p_E = \frac{\varepsilon_0}{2} E_n^2. \tag{41}$$

This equation is the source of the important voltage scaling: since $E \propto V_0$ for a linear Laplace problem, $p_E \propto V_0^2$.

4.8 Interface Normal and Curvature

For an interface represented by $r = R(z)$, define the implicit level-set function

$$F(r, z) = r - R(z). \tag{42}$$

A unit normal pointing toward increasing r is

$$\mathbf{n} = \frac{\nabla F}{|\nabla F|} = \frac{(1, -R_z)}{\sqrt{1 + R_z^2}}. \tag{43}$$

The normal electric field sampled at the interface is

$$E_n = \mathbf{E} \cdot \mathbf{n} = \frac{E_r - R_z E_z}{\sqrt{1 + R_z^2}}. \tag{44}$$

Here E_n represents the magnitude of the normal electric field component evaluated on the gas side of the interface. Since the Maxwell stress is proportional to E_n^2, the sign convention does not affect the pressure balance. The axisymmetric curvature of the surface of revolution is

$$\kappa = \frac{1}{R\sqrt{1 + R_z^2}} - \frac{R_{zz}}{(1 + R_z^2)^{3/2}}. \tag{45}$$

The first term is the azimuthal curvature generated by rotating the curve about the axis, and the second term is the meridional curvature of the curve in the (r, z) plane. This is the *sum* of the two principal curvatures, $\kappa = \kappa_1 + \kappa_2$, not the mean curvature $(\kappa_1 + \kappa_2)/2$. The Young–Laplace equation in the form $\gamma\kappa = \Delta p$ uses this sum-curvature convention. This expression is used directly in the residual computation.

4.9 Young–Laplace–Maxwell Residual

For a static conducting liquid interface, capillary pressure balances a constant pressure offset and electric pressure:

$$\gamma\kappa = \Delta p + \frac{\varepsilon_0}{2} E_n^2. \tag{46}$$

Rearranging gives the pointwise residual

$$\mathcal{R}(z) = \gamma\kappa(z) - \Delta p - \frac{\varepsilon_0}{2} E_n(z)^2. \tag{47}$$

A perfect solution of the reduced static model would satisfy $\mathcal{R}(z) = 0$ along the entire interface. In practice, the solver evaluates a candidate interface and reports the RMS residual

$$\mathcal{R}_{\mathrm{RMS}} = \left(\frac{1}{N} \sum_{m=1}^{N} \mathcal{R}(z_m)^2 \right)^{1/2}. \tag{48}$$

Since Δp is an unknown constant pressure offset, the implementation removes its mean contribution by choosing

$$\Delta p = \frac{1}{N} \sum_{m=1}^{N} \left(\gamma\kappa(z_m) - \frac{\varepsilon_0}{2} E_n(z_m)^2 \right). \tag{49}$$

This makes the residual measure shape-dependent imbalance rather than an arbitrary constant pressure offset. The residual is therefore a diagnostic of physical balance, not merely a numerical iteration error.

4.10 Reduced Shape Optimization

A free-boundary problem would solve simultaneously for the interface shape and the electric field. The current solver uses a lower-dimensional approximation: the interface is restricted to an analytic cone-family shape with a small number of shape parameters, such as half-angle and apex radius. The optimizer minimizes an objective of the form

$$J(\mathbf{a}) = \mathcal{R}_{\mathrm{RMS}}(\mathbf{a}) + \lambda\, P(\mathbf{a}), \tag{50}$$

where $\mathbf{a}$ denotes shape parameters and P penalizes invalid or off-grid geometries if needed. This formulation keeps the problem computationally light and interpretable. Its limitation is that grid-mask boundary representation can shift the optimized angle away from the ideal Taylor value, which is why the optimized-angle error is treated as a known numerical limitation.

4.11 Effective Space-Charge Shielding

Poisson's equation requires a closure for ρ_e. The project uses a hierarchy of effective closures rather than claiming a complete microscopic emission model. The Gaussian closure is

$$\rho_e(r,z) = \rho_0 \exp\left[-\frac{(r-r_a)^2 + (z-z_a)^2}{2\ell^2}\right], \tag{51}$$

where (r_a, z_a) is the charge-cloud center and ℓ is the cloud width. This closure is useful for controlled numerical experiments because the location, sign, and magnitude of charge are explicit.

The threshold closure activates charge only in high-field regions:

$$\rho_e = \rho_{\max}\left[1 - \exp\left(-\frac{|\mathbf{E}| - E_c}{E_s}\right)\right]_+, \tag{52}$$

where E_c is a critical field scale, E_s controls smoothness, and $\rho_{\max}$ is a saturation density. The notation $[x]_+ = \max(x, 0)$ ensures that nodes with $|\mathbf{E}| < E_c$ produce zero effective charge.

Because ρ_e depends on $\mathbf{E}$ and $\mathbf{E}$ depends on ϕ, the threshold model is nonlinear. The solver uses fixed-point iteration:

$$\nabla^2 \phi^{(k+1)} = -\rho_e^{(k)}/\varepsilon_0, \tag{53}$$

$$\mathbf{E}^{(k+1)} = -\nabla \phi^{(k+1)}, \tag{54}$$

$$\widetilde{\rho}_e^{(k+1)} = F(|\mathbf{E}^{(k+1)}|), \tag{55}$$

$$\rho_e^{(k+1)} = (1-\omega)\rho_e^{(k)} + \omega \widetilde{\rho}_e^{(k+1)}, \tag{56}$$

where $0 < \omega \leq 1$ is an under-relaxation parameter. The iteration stops when the relative change in ρ_e is below a prescribed tolerance or when a maximum iteration count is reached.

4.12 Shielding Metric

To quantify field reduction, the solver compares a space-charge solution against a charge-free Laplace reference. A local region-of-interest metric is

$$S_E = 1 - \frac{\max_{\Omega_{\mathrm{ROI}}} |\mathbf{E}_{\mathrm{shielded}}|}{\max_{\Omega_{\mathrm{ROI}}} |\mathbf{E}_{\mathrm{Laplace}}|}. \tag{57}$$

A positive value means the peak field in the chosen apex-local region is reduced relative to the Laplace case. A negative value indicates either field enhancement, sign sensitivity, or that the chosen region is dominated by boundary artifacts. The region of interest is important because global peak fields may occur at electrode corners or artificial computational boundaries rather than at the cone apex.

4.13 Nondimensional Interpretation

Let L be a characteristic length scale and V_0 the applied voltage scale. Define

$$\hat{r} = \frac{r}{L}, \qquad \hat{z} = \frac{z}{L}, \qquad \hat{\phi} = \frac{\phi}{V_0}. \tag{58}$$

The electric field scale is

$$E_0 = \frac{V_0}{L}. \tag{59}$$

The electric pressure scale is

$$p_{E,0} = \varepsilon_0 \frac{V_0^2}{L^2}, \tag{60}$$

while the capillary pressure scale is

$$p_{\gamma,0} = \frac{\gamma}{L}. \tag{61}$$

Their ratio is the electric Bond number

$$Bo_E = \frac{\varepsilon_0 V_0^2}{\gamma L}. \tag{62}$$

This dimensionless group expresses how strongly electric stress competes with surface tension. Increasing voltage raises Bo_E, while increasing surface tension lowers it. This relationship motivates experimental onset-voltage trends: larger surface tension or larger characteristic length scale generally requires higher voltage for comparable electric-capillary balance, although real electrospray onset also depends on conductivity, flow, electrode shape, and emission physics.

A dimensionless charge density can be defined as

$$\hat{\rho}_e = \frac{L^2 \rho_e}{\varepsilon_0 V_0}, \tag{63}$$

so the nondimensional Poisson equation becomes

$$\hat{\nabla}^2 \hat{\phi} = -\hat{\rho}_e. \tag{64}$$

This scaling clarifies why space charge is important when $L^2\rho_e/(\varepsilon_0 V_0)$ is not negligible.

4.14 Model-Based Onset Criterion

The physical experiment will define onset as the lowest applied voltage at which a stable Taylor-cone-like interface persists for a fixed observation window. A simulation cannot observe stability directly unless it includes time-dependent fluid dynamics, so the solver must use a reduced static criterion. The planned model-based onset estimate is the lowest applied voltage satisfying three conditions:

1. the residual-minimizing interface is Taylor-cone-like and remains inside the computational geometry;

2. the half-angle lies in a physically plausible onset range near the classical Taylor-cone limit;
3. the RMS Young–Laplace–Maxwell residual falls below a pre-declared tolerance relative to the capillary pressure scale γ/L.

Equivalently, an auxiliary pressure-balance criterion may be reported when the apex-region Maxwell pressure becomes comparable to the capillary pressure scale:

$$\frac{\varepsilon_0 E_{n,\mathrm{apex}}^2/2}{\gamma/L} = O(1). \tag{65}$$

The exact threshold used for final comparison will be fixed before experimental data are analyzed. This prevents tuning the numerical onset criterion after seeing the measurements.

4.15 Summary of Theoretical Deliverables

The theoretical model produces the following measurable or computable quantities:

- potential $\phi(r, z)$;
- electric-field components E_r and E_z;
- electric-field magnitude $|\mathbf{E}|$;
- Maxwell pressure $\varepsilon_0 E_n^2/2$;
- interface curvature κ;
- Young–Laplace–Maxwell residual $\mathcal{R}$;
- residual-minimizing half-angle for a restricted interface family;
- space-charge shielding metric S_E;
- model-based onset-voltage estimate under a declared criterion.

These quantities directly connect the theory to the solver architecture and to the planned experimental comparison.

5 Solver Architecture

5.1 Implementation Overview

The solver is implemented as a modular Python package. Its main modules are summarized in Table 1. The design goal is to keep the numerical core transparent and testable while using sparse linear algebra for consumer-hardware feasibility.

Table 1: Main solver modules.

Module	Responsibility
`grid.py`	Axisymmetric grid and indexing convention
`operators.py`	Sparse Laplace/Poisson finite-difference operator
`boundary_conditions.py`	Dirichlet boundary insertion
`electrostatics.py`	Laplace/Poisson solve entry points
`fields.py`	Electric-field reconstruction and interpolation
`geometry.py`	ImplicitCone with C1 rounded cap-flank geometry
`immersed.py`	Fractional-distance immersed Dirichlet stencils
`interface.py`	Interface geometry, curvature, half-angle extraction
`residual.py`	Young–Laplace–Maxwell residual diagnostics
`space_charge.py`	Gaussian and threshold space-charge closures
`optimization.py`	Cone-family shape optimization with onset projection
`verification.py`	Immersed verification workflows
`app_backend.py`	Streamlit-independent backend for interactive app
`validation_campaign.py`	Angle scans and numerical-sensitivity analysis
`fem_reference.py`	Independent Q1 FEM for Poisson benchmarks
`fitted_fem.py`	Body-fitted P1 FEM for prescribed rounded cones
`fixed_cone_validation.py`	Common FD–FEM field and stress diagnostics
`image_processing.py`	Image preprocessing and contour extraction
`profile_extraction.py`	Pixel-to-profile conversion and half-angle fitting
`synthetic_validation.py`	Synthetic image rendering and extraction tests
`comparison.py`	Profile alignment and geometric error metrics
`run_validation.py`	Public-data extraction and fixed-pressure checks

The nine modules in the lower block belong to the analysis campaign summarized in Section 6. They are not part of the `solver/` package distributed with this repository, so the campaign results reported in Sections 6 and 7 cannot be regenerated from the accompanying code as it stands; only the regular manufactured Poisson convergence is reproduced there.

5.2 Computational Pipeline

A typical solver run follows this sequence:

1. Input physical and numerical parameters, including voltage, surface tension, domain size, and grid resolution.
2. Build a two-dimensional axisymmetric (r, z) grid.
3. Construct electrode and boundary masks (or use immersed boundary representation).
4. Assemble the sparse axisymmetric Laplace/Poisson matrix.
5. Apply boundary conditions (staircase or immersed Dirichlet stencils) and solve $A\boldsymbol{\phi} = \boldsymbol{b}$.
6. Reconstruct E_r, E_z, and $|\mathbf{E}|$.
7. Sample a candidate interface $R(z)$ from the ImplicitCone geometry.
8. Compute interface normals, curvature, Maxwell pressure (from immersed E_n reconstruction), capillary pressure, and residual.
9. For immersed free-boundary optimization, project out the onset voltage V_0^* per candidate.
10. Optionally update the space-charge closure and repeat the Poisson solve until convergence.

11. Return diagnostics, plots, and exported arrays.

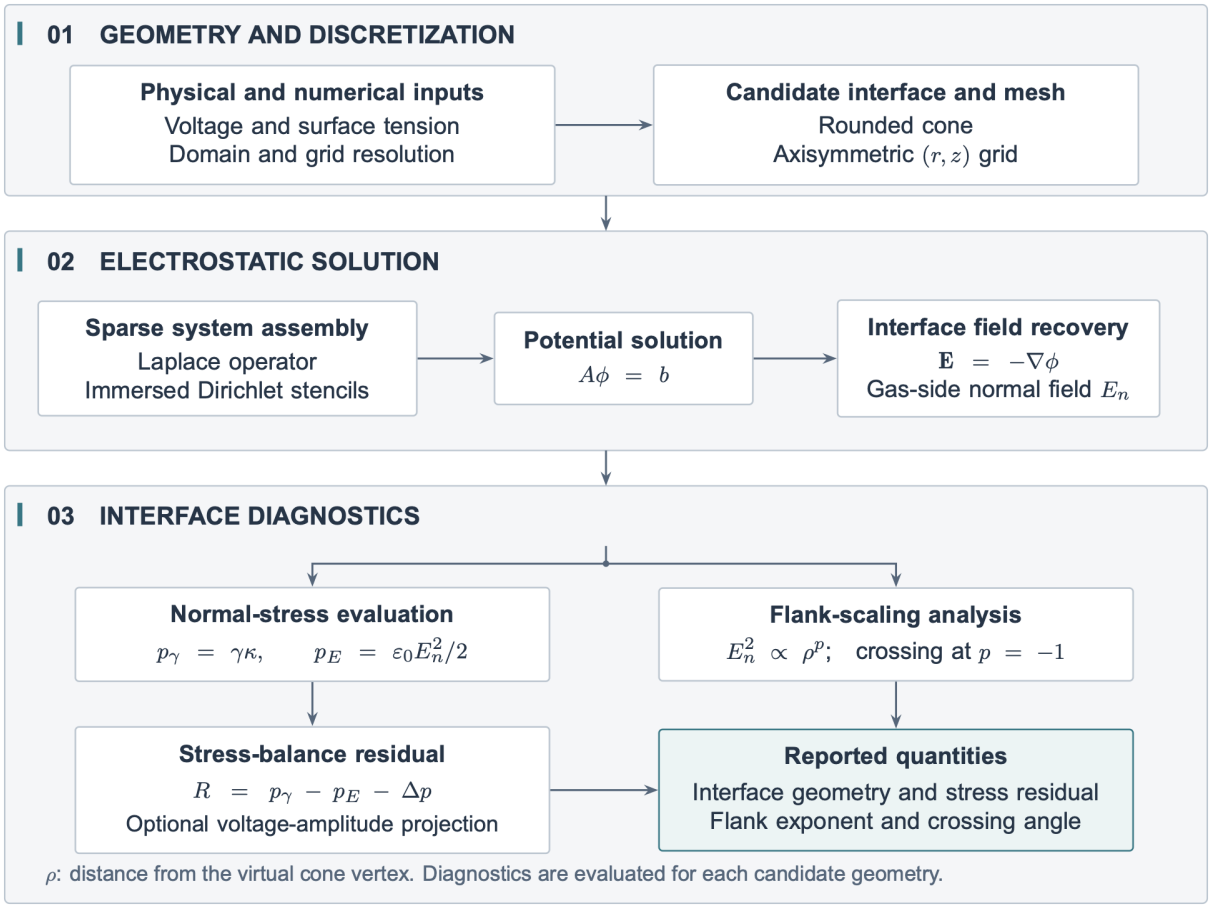


Figure 2: Computational workflow for the charge-free immersed formulation. Geometry and discretization are followed by the electrostatic solution and two interface diagnostics: stress-balance residuals and flank-field scaling. Voltage-amplitude projection is optional.

5.3 Consumer-Hardware Design

The solver uses sparse matrices rather than dense matrices. For a grid of $n_r \times n_z$ nodes, the number of unknowns scales as $n_r n_z$, while the finite-difference operator has only a small number of nonzero stencil entries per row. This allows small-to-moderate grids to run quickly on a standard laptop.

Computational Performance. Solution time scales approximately as $\mathcal{O}(N^{2.3})$ where N is the total number of grid points. For the present implementation, a production-quality 121×177 grid (21,417 points) solves in approximately 13 seconds on the recorded 2020 MacBook Pro test system. This is a project-specific benchmark, not a general claim about all multiphysics software or hardware.

5.4 Interactive App

The repository includes a Streamlit application that exposes basic parameters such as voltage, surface tension, electrode spacing, domain radius, grid resolution, and space-charge model. The

app has two tabs: a *Classic* tab for the legacy staircase boundary representation, and an *Immersed verification* tab that demonstrates the grounded-box onset-projected optimization and the imposed-Taylor identity verification. It returns scalar diagnostics and Plotly visualizations of potential, electric-field magnitude, space-charge density, interface overlay, and residual profile. The app is a demonstration and analysis interface; the scientific contribution remains the solver and validation workflow.

6 Numerical Verification and Sensitivity

The numerical assessment examines the discretization, treatment of curved boundaries, reconstruction of interface fields, and parameter sensitivity. A manufactured solution provides an exact reference for the regular operator, while an independent finite-element discretization is used to assess the curved-interface solution. Synthetic images with known geometries provide a separate evaluation of the measurement procedure.

Provenance. Of the results below, only the regular manufactured Poisson convergence in the first subsection is reproduced by the code distributed with this repository. The curved-boundary, independent finite-element, sensitivity, optimizer, and synthetic-image results were produced by the analysis campaign summarized in Section 5; that campaign's scripts and raw data are not part of this repository and were not rerun here, so those numbers are reported as supplied evidence rather than as independently reproduced verification. Section 8 records the corresponding qualification gap for the image pipeline.

6.1 Regular manufactured Poisson problem

The source-derived quartic test uses $\phi = r^4 + z^4 + r^2 z^2$ and the axisymmetric source $18r^2 + 16z^2$. On grids 17×19, 33×37, 65×73, and 129×145, recorded RMS errors were $1.82155502 \times 10^{-3}$, $4.44529637 \times 10^{-4}$, $1.09730635 \times 10^{-4}$, and $2.72549081 \times 10^{-5}$. Consecutive observed orders were 2.03482, 2.01831, and 2.00938.

Discretization error is evaluated as the RMS difference between the numerical and exact potentials at interior nodes. For halved mesh spacings, the observed order is $q = \log(e_h/e_{h/2})/\log 2$. As shown in Figure 3, the FD error decreases systematically with refinement, and the observed order approaches two. An independently assembled bilinear Q1 finite-element discretization exhibits the same asymptotic behavior. These results are consistent with second-order accuracy for the regular axisymmetric Poisson operator.

6.2 Curved-boundary refinement

The effect of curved-boundary representation is examined by comparing immersed and staircase discretizations on grids of 31×45, 45×65, 61×89, and 87×129. To maintain consistent spatial sampling, the potential is evaluated at the same 56 gas-side locations on all grids. The change between successive levels is defined by

$$D_\ell = \left[\sum_{j=1}^{56} \left(\phi_\ell(\mathbf{x}_j) - \phi_{\ell-1}(\mathbf{x}_j) \right)^2 \right]^{1/2} .$$

Figure 4 shows that D_ℓ decreases from 10.334 to 4.264 and 2.549 V for the immersed discretization. The corresponding staircase values are 25.731, 7.107, and 11.267 V. Apparent orders calculated from

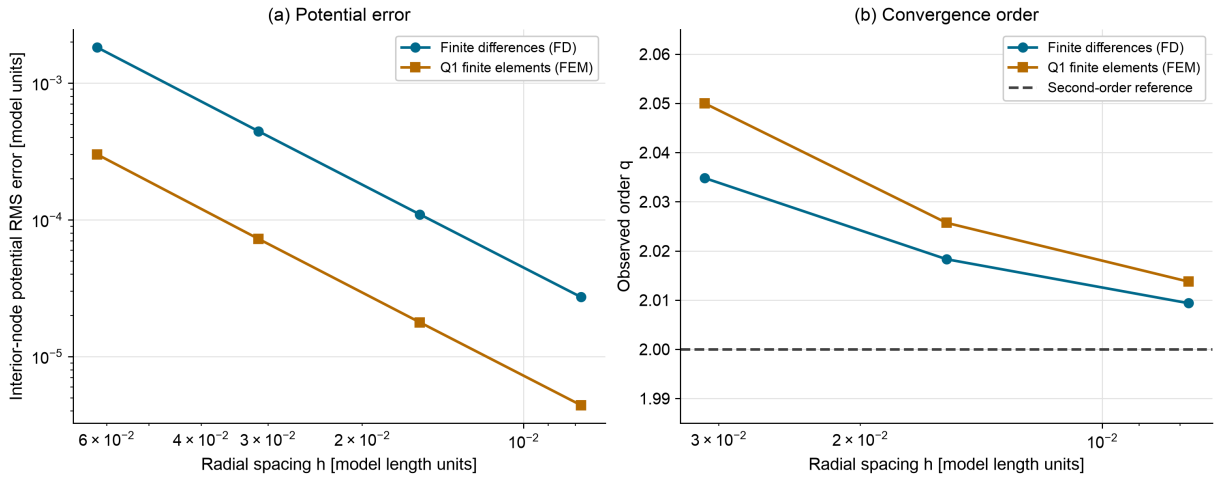


Figure 3: Regular-grid convergence. Errors are measured against the exact manufactured potential; h is radial spacing in model length units.

these changes are 2.854 and 1.429 for the immersed treatment, compared with 4.148 and −1.280 for the staircase treatment. The immersed formulation therefore exhibits a consistent decrease in inter-level variation over the tested grids, whereas the staircase representation produces a reversal at the finest pair of levels. This comparison characterizes the refinement sensitivity of the curved-boundary solution.

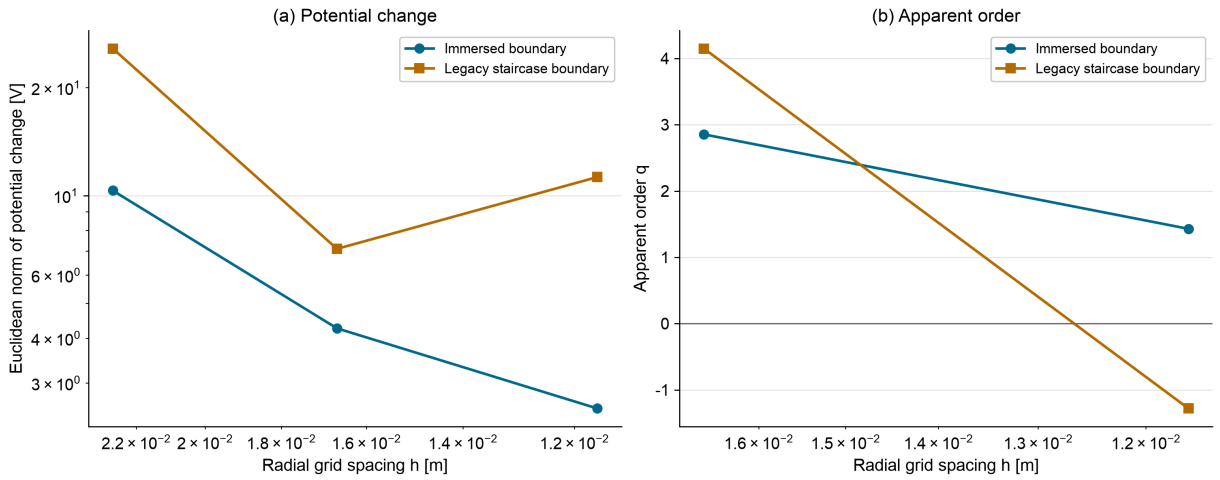


Figure 4: Curved-boundary refinement. Potential changes use fixed gas-side points; apparent orders are calculated from consecutive changes.

6.3 Independent fixed-cone finite-element comparison

An independent comparison is performed between the immersed finite-difference (FD) solver and a body-fitted, piecewise-linear P1 finite-element (FEM) discretization of the same charge-free electrostatic problem. The domain radius and axial extent are 12.5 and 10 mm, respectively. The rounded cone has a half-angle of 49.29°, an apex at $z = 8.6$ mm, and a cap radius of 50 μm. A potential of 1000 V is prescribed on the cone, with the outer boundary grounded. Three FD grids, 65×97,

129×193, and 257×385, are compared with successively refined FEM meshes. Both solutions are evaluated at common gas-side locations and at 41 interface positions over $z = 3$–6 mm.

The finest-level relative RMS differences are 0.024% for potential, 0.226% for the vector electric field, 0.381% for the interface-normal field, and 0.817% for Maxwell pressure $p_E = \varepsilon_0 E_n^2/2$. The projected-voltage difference is 1.071%, and the projected-residual difference is 2.518% when normalized by the RMS of the centered capillary pressure. Identical curvature, arclength weights, pressure projection, and voltage-amplitude definitions are used in the two calculations. The agreement in Figure 5 consequently extends from the primary potential solution to the derived interface stresses and projected residuals.

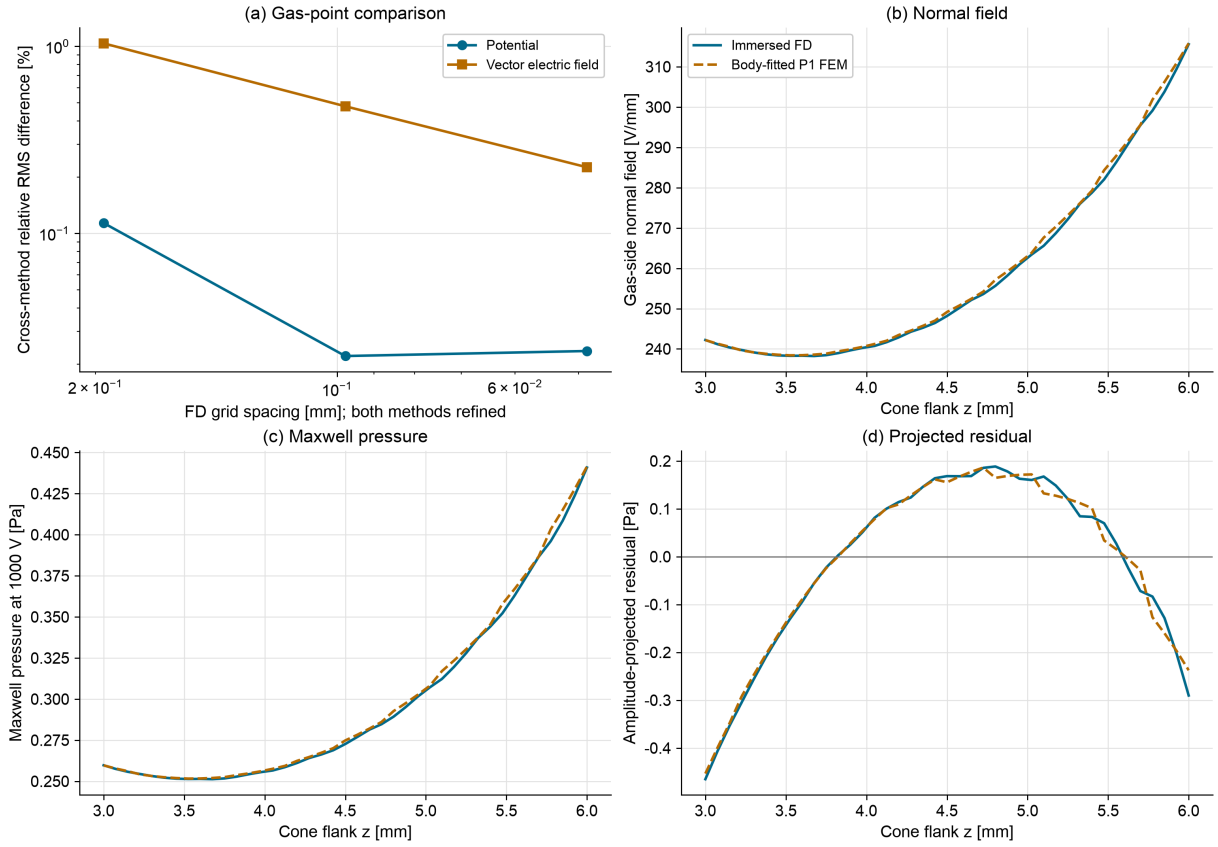


Figure 5: Fixed-cone FD–FEM comparison. (a) Gas-point differences. (b) Interface-normal field. (c) Maxwell pressure at 1000 V. (d) Amplitude-projected residual.

6.4 Angle and parameter sensitivity

The sensitivity of the angle diagnostics is assessed over candidate half-angles of 46°–52° by varying mesh resolution, domain size, cap size, flank-sampling window, and outer-boundary condition. Figure 6 compares the projected-residual landscape with the exponent obtained by fitting $E_n^2 \propto \rho^p$, where ρ denotes distance from the virtual cone vertex. Under the analytical Taylor far-field condition, the residual-minimizing angle varies from 48.0° to 50.0°, whereas the $p = -1$ crossing remains within 49.067°–49.176°. Replacing this boundary condition with a grounded outer boundary increases the residual and shifts the exponent curve. The two diagnostics therefore resolve different features: the crossing measures consistency with the imposed Taylor scaling, while the residual

minimum reflects the finite-domain stress balance.

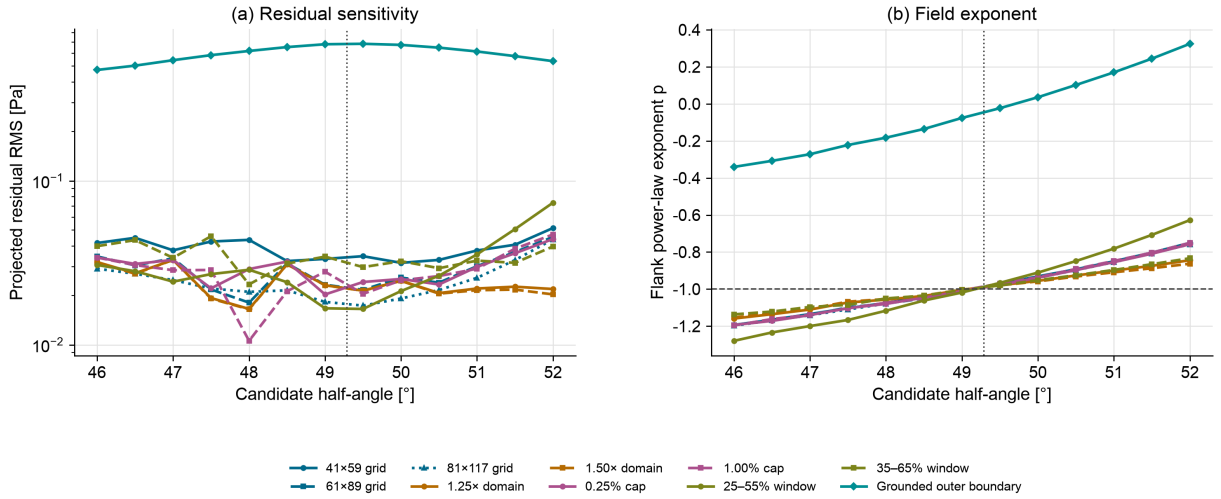


Figure 6: Angle sensitivity. Residual minima and field-exponent crossings respond differently to numerical settings.

Initialization sensitivity is further examined using five angle–apex-radius starting pairs for Powell optimization on the 31×45 grid. The final half-angles span 40.199°–42.791°, and the apex-radius parameters range from 0.0704 to 0.0887 m. Despite these distinct endpoints, the projected RMS residuals remain within 0.01257–0.01333 Pa (Figure 7). This weak variation in objective value indicates an angle–radius ambiguity in the fitted parameterization. Accordingly, the residual magnitude and the recovered geometry provide complementary information on the optimization result.

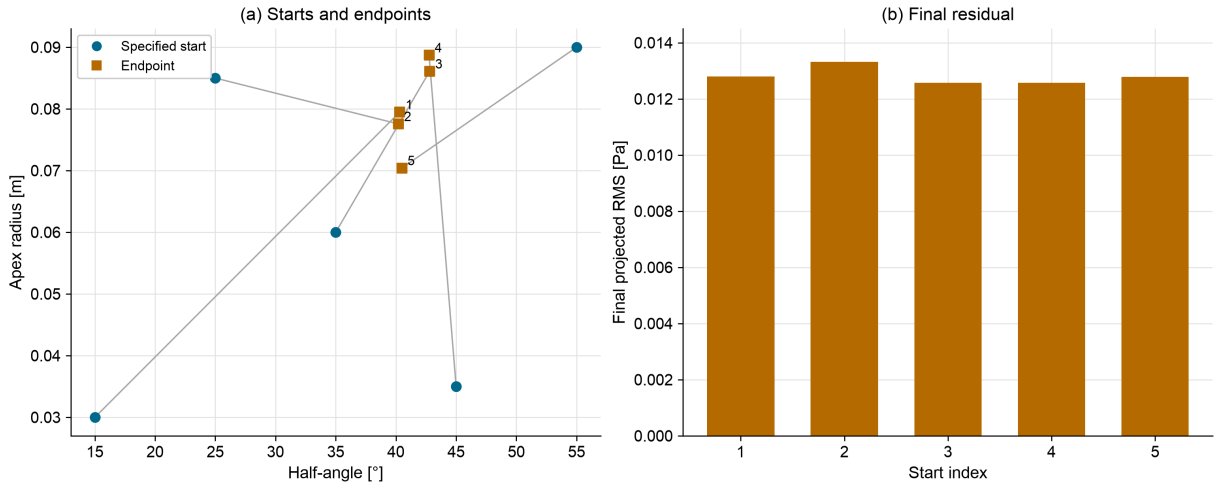


Figure 7: Multistart optimization. Specified starts and final parameter pairs are paired with their final projected residuals.

6.5 Synthetic-image measurement test

The accuracy of image-based geometry extraction is assessed using 108 analytic cone silhouettes. The test combines four half-angles (40°, 45°, 49.29°, and 55°), three visible lengths (100, 200, and 400 pixels), three registered rotations (−5°, 0°, and 5°), and three paired noise–blur settings. Signal-to-noise ratios of 60, 50, and 40 dB are paired with Gaussian blur widths of 1.0, 1.5, and 2.0 pixels, respectively. Image scale and axis rotation are prescribed as calibration data, and the noise realization uses random seed 20260922.

The absolute angle error is defined as $|\alpha_{\text{extracted}} - \alpha_{\text{true}}|$, and the profile nRMSE is the RMS radial difference from the analytic profile normalized by the reference visible cone length. As shown in Figure 8, the campaign reports maximum errors of 0.299° and 1.406%, with all 108 cases below the prescribed 1° and 2% criteria. This reported performance does *not* transfer to the accompanying code. Rerunning the in-repository pipeline (with its declared imaging dependency installed) over three half-angles and four noise levels returned absolute angle errors of 25°–80°, with the acceptance flag false in all twelve cases; the in-repository acceptance predicate also returned a pass when every case raised an exception, which has since been corrected. The synthetic qualification of the measurement procedure is therefore treated as open rather than established, and the public-image results in Section 7 are reported as published-image comparisons, not as a qualified absolute measurement. The reproduced run is recorded in `paper/paper_submission/data/synthetic_validation_results.txt`.

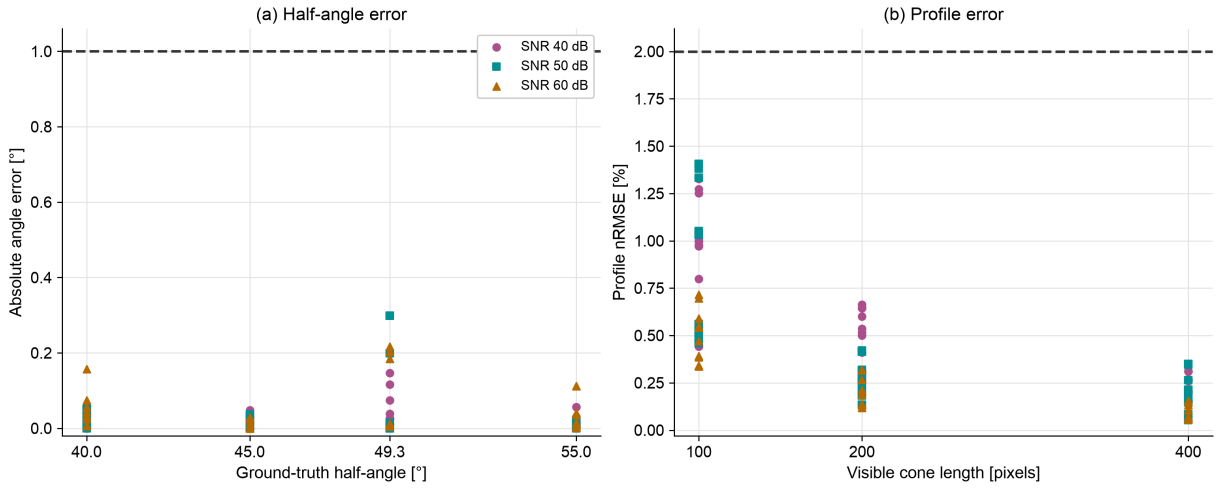


Figure 8: Synthetic-image accuracy. Each marker is one analytic-image case; dashed lines show the preset acceptance limits.

7 Public Experimental Data and Balance Diagnosis

Scope. This section analyses previously published ethanol cone-jet images and reported angles; no new laboratory measurements are presented, and the images and reported series belong to the source study. The extraction and balance numbers below were produced by the analysis campaign summarized in Section 5 and are not reproduced by the code distributed with this repository. Because the synthetic qualification of the extraction pipeline remains open (Sections 6 and 8), the angle comparison is presented as a consistency check between two published image-derived series rather than as validation of the solver.

7.1 Source and experimental conditions

The framework is applied to the ethanol cone-jet photographs and angle–voltage measurements reported by Yang et al. [8]. The photographs are obtained from Figure 5 of that study, and the full cone angles are digitized from Figure 6 and converted to half-angles using $\alpha = \theta/2$. The selected conditions correspond to a flow rate of $Q = 0.10$ μL/s, a nozzle-to-plate spacing of $H = 2$ mm, and an aperture diameter of $D = 2$ mm. The experimental inputs and numerical parameters are summarized in Table 2.

Table 2: Experimental inputs and numerical settings.

Quantity	Value	Source/status
Nozzle outer / inner diameter	0.30 / 0.16 mm	Yang et al., Sec. 1
Plate outer / aperture diameter	80 / 2 mm	Yang et al., Sec. 1; Fig. 6
Nozzle-to-plate spacing	2 mm	Yang et al., Fig. 6
Ethanol flow rate	0.10 μL/s	Yang et al., Figs. 5–6
Temperature; surface tension	25°C; 0.022 N/m	Yang et al., Table 1
Image scale bar	300 μm / 191 pixels	Yang et al., Fig. 5
Outer-domain radius	80 mm	Numerical setting
Exposed needle length	5 mm	Numerical setting
Tip radius; local cell size	6; 4 μm	Numerical setting
Pressure calibration voltage	1.7203 kV	Lowest reported point

The experimental geometry is represented in an axisymmetric (r, z) domain, with the nozzle exit at $z = 0$ and the grounded annular plate at $z = 2$ mm. The radial coordinate r is measured from the symmetry axis. The aperture and plate outer radii are 1 and 40 mm, respectively, as illustrated in Figure 9. The liquid surface and nozzle are held at the applied experimental voltage. The exposed needle length, tip regularization, and outer-domain dimensions are prescribed as numerical parameters.

7.2 Photograph measurement

Four photographs acquired at 1.80, 1.95, 2.10, and 2.30 kV are analyzed using a common spatial calibration of $300/191 = 1.57068$ μm/pixel. A grayscale threshold of 110 is used for the nominal extraction. The paired left and right boundaries determine the registered axis, and linear fits over 30–75% of the visible height yield the cone half-angle. The fitted virtual tip is used for profile alignment, whereas the visible height is measured directly from the nozzle exit to the observed tip. Figure 10 presents the extracted contours and registered axes superimposed on the source photographs.

As the voltage increases from 1.80 to 2.30 kV, the extracted half-angle increases from 22.26° to 41.64°, while the visible height decreases from 301.6 to 147.6 μm (Table 3). The registered profiles in Figure 11 show the associated widening and axial shortening of the observed cone. Presenting the profiles on a common physical scale allows these geometric changes to be compared directly.

7.3 Comparison with reported angles

Quantitative comparison is restricted to the three voltage pairs whose mismatch is below 0.005 kV: 1.80, 1.95, and 2.10 kV. The photo-derived and reported half-angles are listed separately in Table 3. Their absolute differences are 1.970°, 1.520°, and 0.522°, respectively, corresponding to a mean

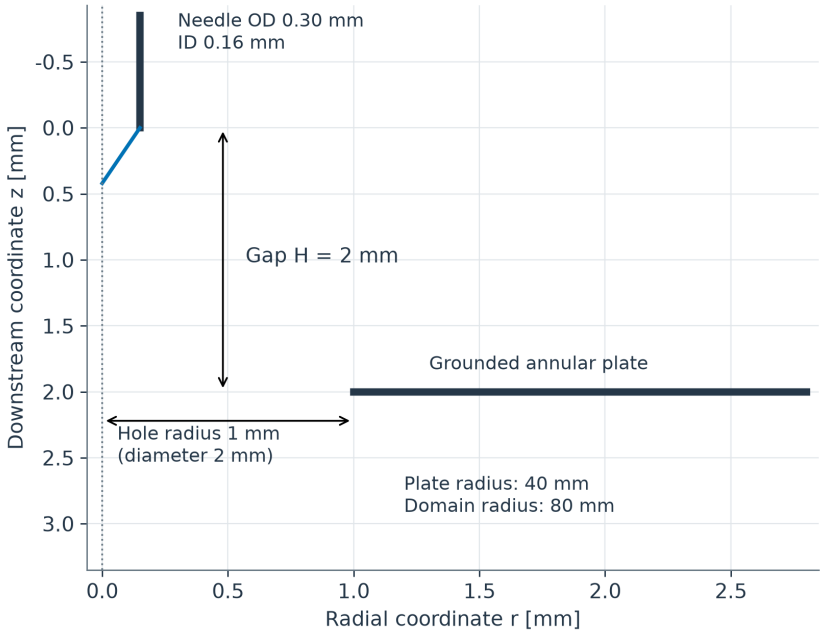


Figure 9: Axisymmetric representation of the experimental case. Plate and aperture dimensions are reported inputs; the outer-domain radius is a numerical setting.

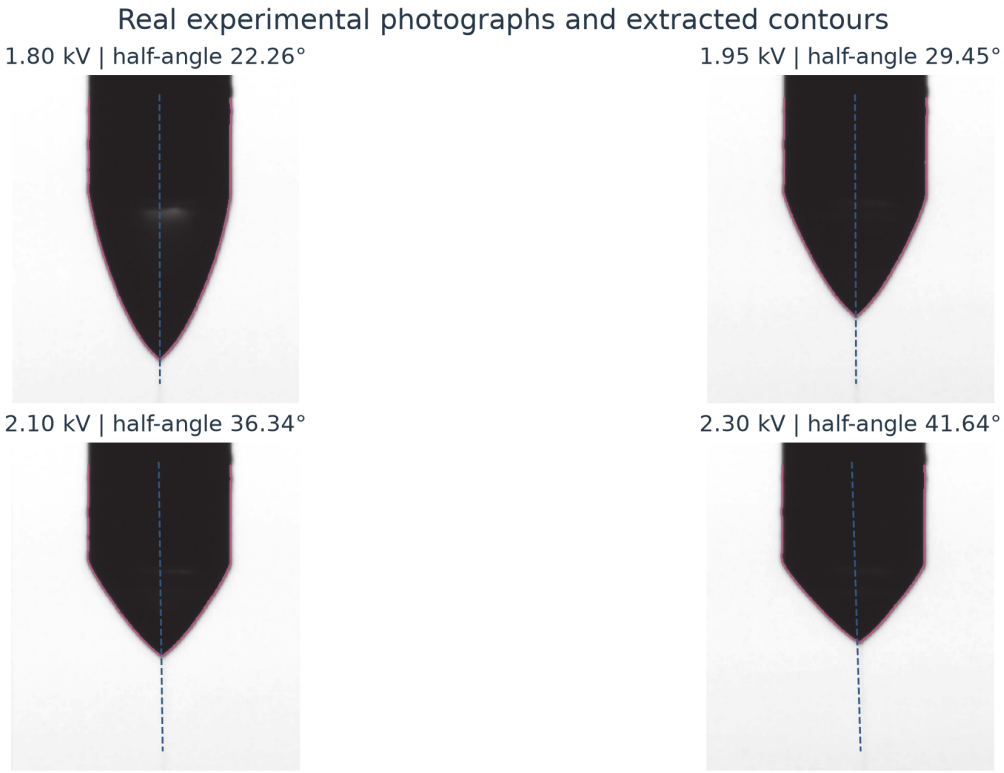


Figure 10: Ethanol photographs with extracted contours. Source: Yang et al., Figure 5; contour and axis overlays are from the present analysis.

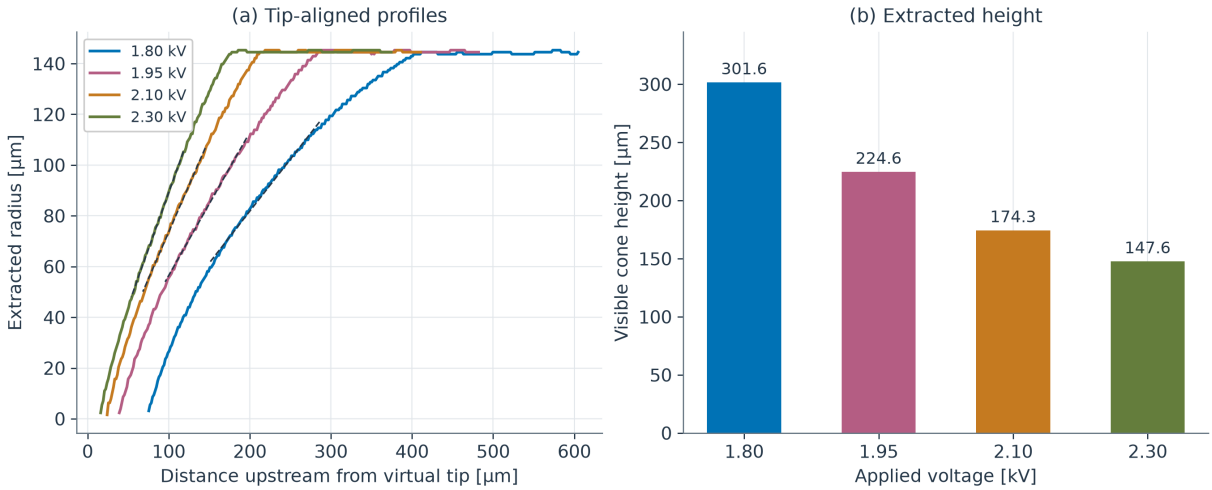


Figure 11: Registered profiles and visible heights. Dashed flank fits summarize the extracted contours.

absolute difference of 1.337° and an RMS difference of 1.468°. Figure 12 shows that both measurement series increase with voltage, with the largest matched discrepancy occurring at 1.80 kV. As a consistency check between two independently published image-derived series, the comparison is informative; because the extraction pipeline has not passed its synthetic qualification, the absolute differences should not be read as a calibrated measurement error.

Table 3: Photo-derived and reported half-angles.

Photo V (kV)	Reported V (kV)	Photo α (°)	Reported α (°)	$\lvert\Delta\alpha\rvert$ (°)	Photo height (μm)
—	1.7203	—	19.80	—	—
1.80	1.8005	22.26	24.23	1.970	301.6
1.95	1.9503	29.45	30.97	1.520	224.6
2.10	2.1000	36.34	36.86	0.522	174.3
—	2.2503	—	42.99	—	—
2.30	—	41.64	—	—	147.6

Dashes indicate an unpaired source observation. The reported series and photographs come from separate figures in the same experimental study.

7.4 Extraction sensitivity

The influence of segmentation is assessed by repeating the extraction at grayscale thresholds of 90, 110, and 130. The maximum half-angle range is 0.204° for the three voltage-matched cases and 0.322° for the 2.30 kV photograph. At the nominal threshold, the RMS distances from the extracted contours to their straight-flank fits are 1.082, 0.837, 0.848, and 0.579 μm in ascending voltage order (Figure 13). The threshold-induced variation is smaller than the matched inter-source angle differences. These results distinguish the sensitivity of the extraction procedure from the geometric fit error and the discrepancy between measurement sources.

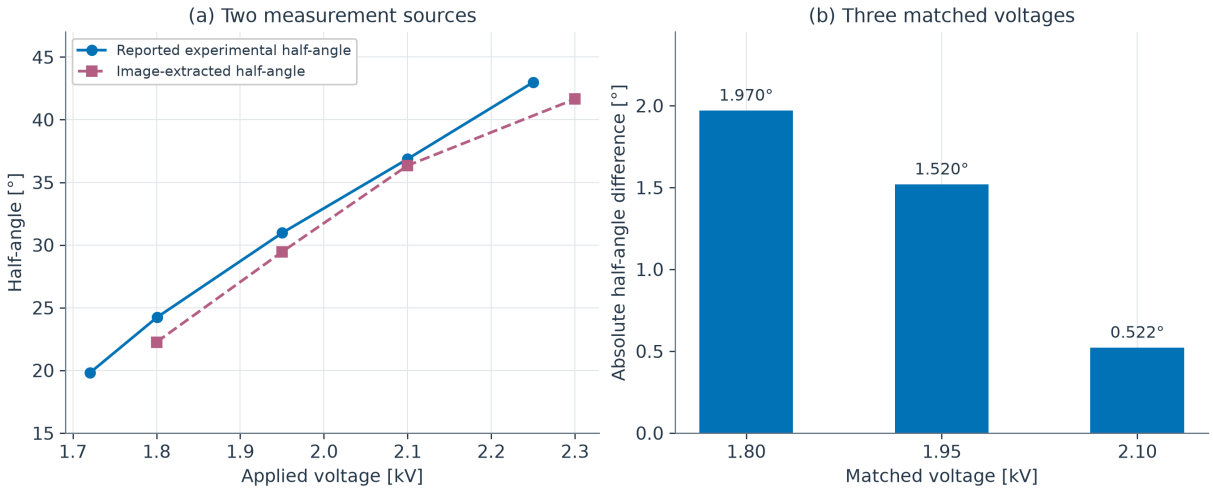


Figure 12: Experimental angle comparison. Photo extraction and reported markers are distinguished; differences use the three matched voltages.

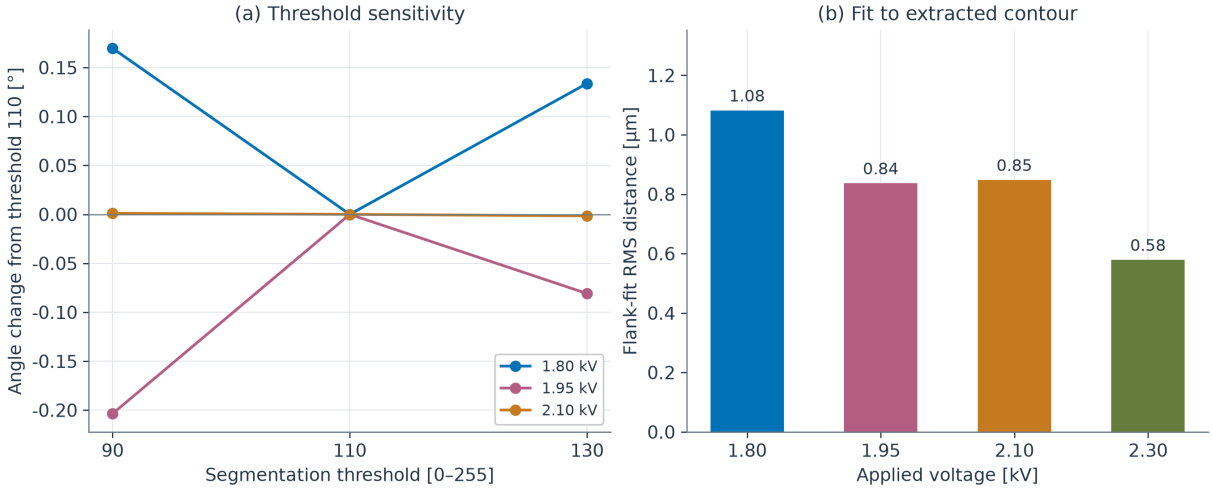


Figure 13: Extraction sensitivity. Threshold-induced angle changes and nominal straight-flank fit distances.

7.5 Fixed-pressure stress-balance diagnosis

The applicability of the constant-pressure balance is examined using the reported half-angle as a geometric input at each voltage. The gas-side electric field is computed at the corresponding applied voltage. Over the central 25–75% of the cone height, the residual and its area-weighted mean are defined as

$$R(z;V) = \gamma\kappa(z) - \frac{\varepsilon_0}{2}E_n(z;V)^2 - \Delta p, \qquad \overline{R} = \langle R \rangle_A.$$

The pressure offset is calibrated at the lowest-voltage condition according to

$$\Delta p = \left\langle \gamma\kappa - \frac{\varepsilon_0}{2}E_n^2 \right\rangle_{A,V=1.7203\,\mathrm{kV}} \simeq 6.458\ \mathrm{Pa},$$

with field values interpolated from the precomputed solutions. This pressure is then held constant over the remaining voltage conditions. The mean residuals are approximately 0, −44, −123, −201, and −278 Pa for the five reported conditions. The increasingly negative residual indicates that the sum of electric pressure and the fixed pressure offset exceeds the capillary pressure of the reconstructed interface. Calibration imposes mean balance at the lowest voltage, while the spatial residual retains the local variation along the interface.

A spherical reference problem is used to assess the accuracy of the normal-field reconstruction. Refining the local spacing from 6 to 2 μm reduces the mean field error from 0.440% to 0.094% and the maximum error from 1.245% to 0.328%. The balance analysis is repeated with a 2 μm spacing, exposed needle lengths of 2 and 10 mm, and a tip radius of 12 μm. For each setting, the pressure is calibrated at the same lowest-voltage condition. As shown in Figure 14, the negative high-voltage residual trend persists across these variations. The combined results indicate a systematic departure from the constant-pressure static balance for the selected observed-angle series over the tested numerical settings.

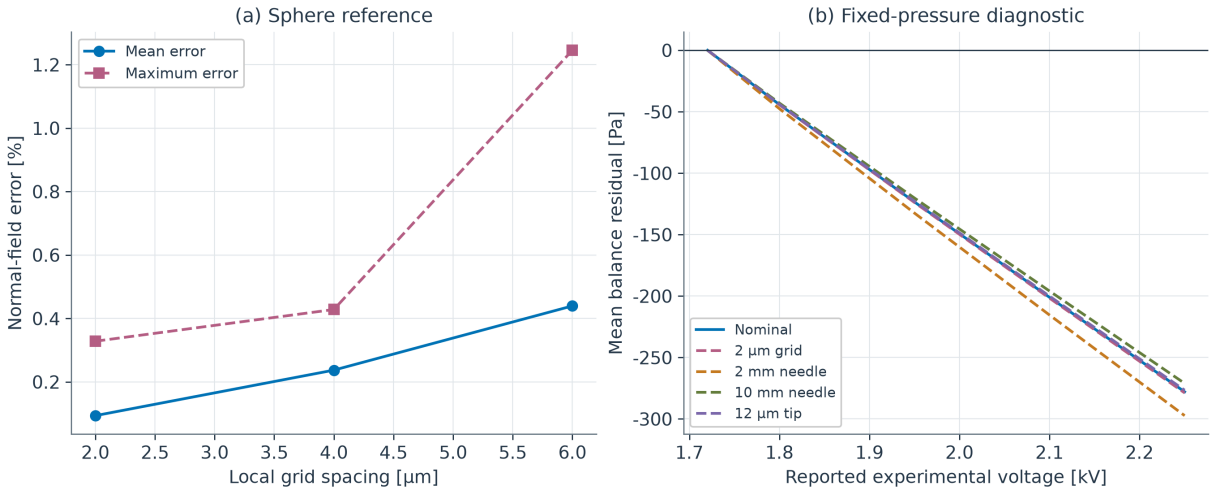


Figure 14: Field and balance checks. (a) Sphere-reference normal-field errors. (b) Mean residuals at the reported angles for five numerical settings.

8 Image Processing and Comparison Method

8.1 Computer-Vision Pipeline

The image-processing pipeline will convert recorded Taylor-cone images into quantitative interface coordinates. The planned workflow is:

1. select stable frames from video sequences,
2. crop the region of interest around the meniscus,
3. calibrate pixels to physical length using a known needle/nozzle dimension or calibration target,
4. apply thresholding and/or edge detection,
5. extract the cone boundary contour,
6. fit local interface slopes near the cone region,
7. compute the experimental half-angle,

8. export $(r_{\text{exp}}, z_{\text{exp}})$ profile coordinates and uncertainty estimates.

Before physical data are collected, the pipeline will be tested on synthetic cone images with known ground-truth angles. The current end-to-end implementation has not passed this qualification: a diagnostic rerun on a nominal 49.29° cone produced large angle and profile errors even at zero added noise. Consequently, this section is a proposed qualification protocol only; it is not evidence that the pipeline, camera calibration, or physical solver is validated. The repair gate requires held-out rendered images, explicit failure counts, fixed coordinate conventions, and predeclared thresholds before any experimental accuracy claim is made.

8.1.1 Frame Selection

Video sequences recorded during voltage ramps will contain hundreds to thousands of frames. Frame selection identifies the subset of frames that satisfy predefined operating-state stability criteria. The selection algorithm will:

1. Divide the video into segments corresponding to each voltage step (using timestamps or manual annotation).
2. For each segment, compute a time series of shape metrics (e.g., apex position, interface curvature, centroid location).
3. Identify the longest contiguous sub-interval within the segment where shape metrics remain within the stability tolerance bands.
4. Select representative frames from the stable interval. Extract each frame separately and report the within-interval variation; do not treat adjacent video frames as independent experimental trials.
5. Use a predeclared summary (for example, the median profile) only after checking that frame-to-frame variation is small relative to the measurement uncertainty.

If no stable interval exists within a voltage segment, that segment will be classified as "unstable" and excluded from quantitative comparison.

8.1.2 Calibration

Pixel-to-millimeter calibration will use one of two methods:

1. **Calibration target method:** A precision-machined scale or printed checkerboard pattern will be placed in the focal plane before the experiment. The pixel spacing corresponding to a known physical distance will be measured using edge detection or corner detection. This method is preferred when the camera-to-object distance is fixed and repeatable.
2. **Needle diameter method:** The outer diameter of the needle will be measured independently using digital calipers or microscopy (e.g., $d_{\text{out}} = 0.80 \pm 0.02$ for a standard 21-gauge needle). In the image, the needle diameter in pixels will be measured by fitting two parallel lines to the needle sidewalls far from the meniscus. The calibration scale is then $\text{mm/pixel} = d_{\text{out}}/d_{\text{pixels}}$.

Calibration uncertainty will be propagated through to all extracted dimensions. If lens distortion is significant, a distortion correction map will be applied using the calibrated-camera procedure described by Zhang [9]. The distortion threshold, if one is used, will be declared from the calibration residuals rather than treated as a universal 1% rule.

8.1.3 Image Preprocessing and Edge Detection

Raw grayscale images will be preprocessed to enhance the liquid-gas interface contrast:

1. **Cropping:** Extract a region of interest (ROI) containing the needle tip and meniscus, discarding background pixels far from the interface.
2. **Histogram equalization or normalization:** Adjust intensity distribution to maximize dynamic range within the ROI.
3. **Gaussian blur (optional):** Apply a small Gaussian kernel (e.g., $\sigma = 1$–2 pixels) to reduce sensor noise, with kernel size chosen to preserve edge sharpness.

Edge detection will use one of the following methods, selected based on performance on synthetic validation images:

- **Canny edge detection:** A two-threshold gradient-based method that produces thin, connected edges. Hysteresis thresholds will be tuned on synthetic images and held fixed for all experimental trials.
- **Otsu thresholding + contour extraction:** Automatically select an intensity threshold that separates liquid (dark) from gas (bright) regions, then extract the boundary contour using a connected-component algorithm. This method is robust to lighting variation but may struggle with low-contrast or noisy images.
- **Sobel gradient magnitude + non-maximum suppression:** Compute the image gradient, threshold the magnitude, and suppress non-local-maximum pixels to produce thin edges. This is a simpler alternative to Canny.

The chosen method will be documented in the final validation report along with all parameter values (e.g., Canny thresholds, Gaussian σ, Otsu method).

8.1.4 Contour Extraction and Smoothing

After edge detection, the interface contour will be extracted as an ordered list of pixel coordinates $\{(x_i, y_i)\}_{i=1}^{N_{\text{contour}}}$. Post-processing steps include:

1. **Contour selection:** If multiple contours are detected (e.g., needle edge, interface, artifacts), select the contour that:
   - originates near the needle tip,
   - extends into the region between the needle and extractor,
   - has monotonically increasing axial coordinate (no backtracking),
   - has the longest arclength among candidates.
2. **Axisymmetry enforcement:** Since the experiment observes a 2D projection of an axisymmetric interface, the pipeline will extract one side of the contour (e.g., left or right boundary) and assume radial symmetry. If both sides are visible, the two half-profiles can be averaged to reduce asymmetry artifacts from optical misalignment.
3. **Smoothing (optional):** Apply a low-pass filter (e.g., Savitzky–Golay filter, cubic spline with smoothing parameter) to reduce pixel-level noise while preserving the overall cone shape. The smoothing strength will be validated on synthetic images to ensure it does not bias the extracted angle.

8.1.5 Coordinate Transformation to (r, z)

The pixel coordinates will be transformed into the solver's axisymmetric coordinate system:

1. Apply the calibration scale: $(x_{\text{mm}}, y_{\text{mm}}) = \text{scale} \times (x_{\text{px}}, y_{\text{px}})$.
2. Identify the needle tip or apex as the coordinate origin. For needle-tip referencing, locate the point where the interface detaches from the needle (sharp curvature change or minimum radius). For apex referencing, locate the point of maximum axial penetration into the gas.
3. Define the axial direction z (parallel to needle axis, positive toward extractor) and the radial direction r (perpendicular to axis, positive away from centerline).
4. Transform the contour into (r, z) coordinates: $r_i = |x_i|$ (distance from axis), $z_i = y_i$ (axial position).

The experimental profile is now represented as $\{(r_i^{\text{exp}}, z_i^{\text{exp}})\}_{i=1}^{N_{\text{exp}}}$.

8.1.6 Half-Angle Measurement

The cone half-angle will be extracted by fitting a straight line to the flank region of the experimental profile. The fitting procedure is:

1. **Define flank region:** Select the interval $z \in [z_{\min}, z_{\max}]$ where the interface is approximately conical. Exclude the apex region (high curvature, rounded cap) and the base region (near needle tip, attachment artifacts). A typical choice is the middle 50% of the visible cone length, e.g., $z_{\min} = z_{\text{apex}} - 0.25L_{\text{cone}}$, $z_{\max} = z_{\text{apex}} - 0.75L_{\text{cone}}$, where L_{cone} is the total cone length from apex to needle.
2. **Least-squares line fit:** Fit $r = mz + b$ to the flank points using weighted or unweighted least squares. The slope $m = \tan(\alpha)$ gives the half-angle:
$$\alpha_{\text{exp}} = \arctan(m). \tag{66}$$
3. **Uncertainty estimate:** The standard error of the slope σ_m from the regression can be propagated to the angle:
$$\sigma_\alpha = \frac{\sigma_m}{1 + m^2}. \tag{67}$$
If multiple frames are analyzed, the angle standard deviation across frames provides an additional uncertainty estimate.

The flank-window selection will be validated on synthetic images to ensure it produces accurate angles for known cone geometries and is not biased by the window choice.

8.2 Profile Alignment

Simulation and experiment will be compared in the same coordinate system. Experimental pixel coordinates will be converted to millimeters and aligned so that the extracted apex or another reproducible reference point matches the simulation coordinate origin. The experimental radius $r_{\text{exp}}(z_i)$ will be interpolated onto the simulation z_i locations, or both profiles will be interpolated onto a common coordinate grid.

Alignment Strategy. Three alignment methods will be considered:

1. **Apex alignment:** Translate the experimental profile so that its apex (point of maximum z or minimum r) coincides with the simulated apex. This is geometrically natural but may be sensitive to apex-region noise or cap artifacts.
2. **Needle-tip alignment:** Translate both profiles so that the needle-tip detachment point (experimentally measured, simulated as the cone base) is at the origin. This is robust to apex

uncertainty but requires clear identification of the detachment point in both experiment and simulation.

3. **Flank-region least-squares alignment:** Minimize the profile RMSE over translations and small rotations, using only the flank region (excluding apex and base). This is robust to localized artifacts but adds degrees of freedom that could artificially reduce error.

The primary method will be apex alignment unless preliminary validation on synthetic images shows it is unreliable. The alignment choice will be frozen before processing experimental data.

Interpolation. After alignment, one profile will be interpolated onto the other's coordinate grid. Linear or cubic spline interpolation will be used. If the experimental profile has higher spatial sampling than the simulation grid, the experimental points will be interpolated onto the simulation z_i locations. If the simulation is finer, the opposite will be done. Points where one profile extends beyond the other (e.g., experimental profile reaches closer to the base) will be excluded from RMSE calculation to avoid extrapolation artifacts.

8.3 Validation Metrics

The primary profile metric will be the root mean square error:

$$\mathrm{RMSE}_R = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(R_{\mathrm{sim}}(z_i) - R_{\mathrm{exp}}(z_i)\right)^2}. \tag{68}$$

A normalized version will divide by a characteristic length L_c, such as electrode spacing or observed cone length:

$$\mathrm{nRMSE}_R = \frac{\mathrm{RMSE}_R}{L_c}. \tag{69}$$

The cone-angle error will be

$$\Delta\alpha = |\alpha_{\mathrm{sim}} - \alpha_{\mathrm{exp}}|. \tag{70}$$

For onset voltage,

$$\epsilon_V = \frac{|V_{\mathrm{sim}} - V_{\mathrm{exp}}|}{V_{\mathrm{exp}}}. \tag{71}$$

Repeated trials will report mean, standard deviation, and uncertainty intervals where possible.

Profile RMSE Decomposition. To diagnose the source of profile mismatch, the radial error can be decomposed into components:

- **Angle-driven error:** If the simulated and experimental half-angles differ by $\Delta\alpha$, the radial profiles will diverge as $\Delta r \approx z\tan(\Delta\alpha)$ along the flank. This component can be estimated by comparing the RMSE from the actual profiles to the RMSE from two cones with half-angles α_{sim} and α_{exp} but otherwise identical geometry.
- **Apex-region error:** Near the apex, the experimental profile may have a finite-radius cap due to finite resolution or physical regularization (surface tension smoothing, finite needle radius), while the simulation may impose a sharper or differently sized cap. The apex-region contribution can be isolated by computing RMSE separately for $z < z_{\mathrm{apex}} - \delta$ (apex window) and $z > z_{\mathrm{apex}} - \delta$ (flank window).
- **Base-region error:** Near the needle attachment, contact-line dynamics, wetting behavior, and needle geometry artifacts may cause deviations not captured by the idealized cone model. Base-region RMSE can be computed separately.

Statistical Significance. Repeated trials at the same condition will produce a distribution of measured angles, onset voltages, and profile RMSEs. Agreement between simulation and experiment will be assessed by comparing the simulation prediction to the experimental mean ± standard deviation (or standard error of the mean if the trial count is small). If the prediction falls within the experimental uncertainty band, agreement is strong. If not, the discrepancy magnitude will be reported as a multiple of the experimental standard deviation (e.g., "simulation predicts $\alpha = 47.5°$, experiment measures $49.2 \pm 0.8°$, discrepancy is 2.1σ").

8.4 Synthetic Validation of the Image Processing Pipeline

Before analyzing experimental data, the entire image-processing pipeline will be validated on synthetic images generated from the solver output. The validation procedure is:

Step 1: Generate Synthetic Images.

1. Run the solver at a known geometry (e.g., electrode spacing $L = 10$, needle diameter $d = 0.8$, surface tension $\gamma = 0.022$, applied voltage $V = 3$).
2. Extract the predicted interface profile $R_{\text{sim}}(z)$ and half-angle α_{sim} (ground truth).
3. Render the profile as a 2D silhouette image using a graphics library (e.g., matplotlib filled contour, PIL polygon rendering, or OpenCV drawing functions).
4. Embed the silhouette in a realistic image background: add a needle representation, set pixel resolution (e.g., 1024×768), apply calibration scale (e.g., 50), render a backlight gradient.
5. Corrupt the synthetic image with realistic noise and blur: add Gaussian intensity noise (SNR 40–60 dB), apply Gaussian blur ($\sigma = 1$–2 pixels), simulate lens distortion if applicable, add pixel quantization (8-bit or 12-bit depth).

Step 2: Process Synthetic Images Through Pipeline.

1. Apply the full image-processing pipeline (edge detection, contour extraction, calibration, angle measurement) to the synthetic image.
2. Extract $\alpha_{\text{extracted}}$ and $R_{\text{extracted}}(z)$ from the pipeline output.
3. Compute the round-trip errors:

$$\Delta\alpha_{\text{synthetic}} = |\alpha_{\text{extracted}} - \alpha_{\text{sim}}|, \tag{72}$$

$$\text{RMSE}_{\text{synthetic}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(R_{\text{extracted}}(z_i) - R_{\text{sim}}(z_i)\right)^2}. \tag{73}$$

Step 3: Acceptance Criteria. The image-processing pipeline will be accepted as valid if the synthetic validation shows:

- $\Delta\alpha_{\text{synthetic}} < 1°$ for cone angles in the range 40° to 55° under moderate noise (SNR ≥ 40 dB),
- $\text{nRMSE}_{\text{synthetic}} < 2\%$ of the cone length,
- Round-trip error does not increase systematically with angle, cone length, or noise level (indicating method robustness).

If this future synthetic qualification fails its predeclared criteria, the edge-detection method, smoothing parameters, or flank-window selection will be revised before experimental images are analyzed. Until that gate passes, no synthetic image result is presented as evidence that the image-analysis method is unbiased, experimentally calibrated, or physically validated.

8.5 Planned Output Figures

The final paper will include: raw image frames, detected-edge overlays, extracted cone profiles, simulated-versus-experimental profile overlays, half-angle comparison plots, onset-voltage comparison plots, and runtime scaling figures.

Representative Image Sequence. A figure will show 3–5 frames from a single voltage ramp: (a) no cone (hemispherical meniscus at low voltage), (b) deformed meniscus (subcritical), (c) stable Taylor cone at onset, (d) cone with incipient jet or spray (post-onset). Each frame will include the detected edge overlay and annotated voltage.

Profile Comparison. For each validated trial, a plot will overlay the experimental profile $R_{\text{exp}}(z)$ (solid line with error band) and the frozen simulation prediction $R_{\text{sim}}(z)$ (dashed line) in the same coordinate system. The half-angles will be annotated with fitted flank lines. Residual error $R_{\text{sim}}(z) - R_{\text{exp}}(z)$ will be plotted as a function of z in a separate panel.

Angle and Voltage Comparison. Scatter plots or bar charts will compare simulated versus experimental half-angles and onset voltages across all trials. Error bars will represent experimental uncertainty (standard deviation across repeated trials). A diagonal line $y = x$ will indicate perfect agreement. The plots will quantify systematic bias (mean offset) and random scatter (RMSE about the diagonal).

Voltage-Sweep Landscape. For at least one trial, a figure will show the experimental regime map (voltage vs qualitative state: no cone, unstable, stable, spray) overlaid with the simulated residual landscape (residual RMS vs voltage). This will illustrate how the numerical onset criterion aligns with the experimental onset observation.

Synthetic Validation Results. A supplementary figure will document the synthetic validation: (a) example synthetic image with ground-truth profile overlaid, (b) extracted profile from pipeline, (c) angle error and profile RMSE as functions of noise level and cone angle, demonstrating pipeline accuracy and robustness.

9 Results Summary

Sections 6 and 7 present the numerical verification campaign and the published experimental-data analysis, respectively. Their principal results are summarized below, together with the boundary of what this repository reproduces: the regular manufactured Poisson convergence is reproduced by the accompanying code, the remaining campaign numbers are supplied externally, and the synthetic-image qualification of the extraction pipeline is still open.

The current submission draft is prepared before physical experiments have been completed. Therefore, experimental result locations are explicitly marked as placeholders rather than fabricated or assumed data.

9.1 Computational Results Already Available

Current completed computational results include:

- classical Taylor half-angle benchmark: 49.2900892921°,

- manufactured Poisson convergence under grid refinement,
- voltage scaling check showing $E \propto V$ and $p_E \propto V^2$,
- residual null sanity check with near-zero RMS residual,
- Gaussian and threshold space-charge closure demonstrations,
- automated test suite passing all current tests.

9.2 Experimental Tables to Complete

Table 4: Placeholder for experimental half-angle comparison.

Trial	V_{exp}	α_{exp}	α_{sim}	$\Delta\alpha$
1	TBD	TBD	TBD	TBD
2	TBD	TBD	TBD	TBD
3	TBD	TBD	TBD	TBD

Table 5: Placeholder for interface-profile and onset-voltage validation.

Condition	Profile RMSE	nRMSE	V_{exp}	ϵ_V
Baseline	TBD	TBD	TBD	TBD
Spacing sweep 1	TBD	TBD	TBD	TBD
Spacing sweep 2	TBD	TBD	TBD	TBD

9.3 Criteria for Interpreting Agreement

Agreement will not be treated as binary proof. Instead, the project will report quantitative error and compare it with experimental uncertainty and model assumptions. Strong agreement would mean small angle error, low normalized profile RMSE, and correct onset-voltage trends across controlled geometry changes. Disagreement will be analyzed to identify whether the dominant cause is experimental uncertainty, finite-grid geometry, simplified space-charge closure, or missing physics.

10 Safety, Ethics, and Compliance

The project contains a computational component and a proposed experimental validation component. The computational component can proceed independently. The physical experiment will not begin until appropriate school approval, local scientific review, risk assessment, and qualified adult supervision are secured.

The proposed experimental validation may involve high voltage and flammable solvent, so it must be treated as a hazardous-device and hazardous-chemical activity. Before purchasing, assembling, or operating any apparatus, the project team will prepare required approval forms, a full risk assessment, safety data sheet documentation, supervision plan, schematic review, operating boundaries, and emergency shutoff procedures. Operation will occur only in a supervised school or laboratory environment, with appropriate enclosure, interlock, current limiting, grounding, ventilation, PPE, and waste-disposal procedures as required by school and competition rules.

No live high-voltage demonstration is planned for a public fair setting. Experimental data, images, videos, and analysis results will be presented instead. The apparatus is intended for microfluidic Taylor-cone observation and image-based validation, not as a projectile device or propulsion demonstration. The project will comply with all applicable Hong Kong selection-fair and Regeneron ISEF-affiliated rules.

11 Limitations

The present solver is intentionally reduced-order. It does not solve full transient Navier–Stokes equations, full Taylor–Melcher leaky-dielectric surface charge conservation, droplet breakup, plume dynamics, or complete thruster performance. The effective space-charge closures are phenomenological models for studying Poisson shielding trends, not predictive emitted-current laws.

The axisymmetric assumption is appropriate for onset-level symmetric geometries but may fail after non-axisymmetric instabilities or downstream jet breakup appear. The finite-difference grid representation introduces geometric error, especially near sharp conical interfaces. The current optimizer-angle deviation from the ideal Taylor half-angle highlights this limitation and motivates improved boundary treatment or explicit grid-convergence quantification.

Experimental validation will also have limitations. Cone-angle extraction depends on image resolution, optical alignment, lighting, threshold selection, calibration accuracy, and frame selection. Onset voltage may depend on fluid impurities, humidity, flow stability, electrode condition, and subjective stability criteria unless carefully standardized. Therefore, final conclusions will be stated in terms of quantified agreement and uncertainty rather than exact prediction.

12 Conclusion and Future Work

This project develops a lightweight sparse Python solver for onset-level Taylor-cone modeling. The current codebase already includes an axisymmetric electrostatic solver, field reconstruction, residual diagnostics, effective space-charge closures, a shape optimizer, a Streamlit app, automated tests, and numerical validation artifacts. It reproduces the classical Taylor half-angle benchmark and passes current numerical sanity checks.

The verification campaign reported in Section 6 supplies an independent finite-element comparison, curved-boundary refinement studies, and parameter-sensitivity and optimizer results, while the public-image analysis of Section 7 reports a mean absolute half-angle difference of 1.337° between two published image-derived series and a systematic negative residual trend under a fixed-pressure closure. Of these, only the regular manufactured Poisson convergence is reproduced by the code distributed with this repository; the remaining numbers are supplied by that campaign and are not independently reproduced here. In particular, an in-repository rerun of the image pipeline did not meet its synthetic acceptance criteria (Sections 6 and 8), so the image-derived angles are presented as a published-image comparison rather than as a qualified measurement. The campaign therefore motivates the planned experimental validation rather than replacing it.

Future work includes improving sharp-interface boundary representation, coupling threshold space charge inside the shape-optimization loop, adding more physically constrained charge/current closures, extending toward leaky-dielectric effects where justified, and comparing against higher-fidelity simulations or published experimental datasets. The intended contribution is not to replace full multiphysics simulation, but to provide a transparent, reproducible, and computationally accessible model for Taylor-cone onset studies.

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A Submission Checklist

Before final submission, the following items should be completed:

- Confirm official Hong Kong selection-fair formatting requirements.
- Replace placeholder author/school information.
- Confirm word/page limits and adjust length if needed.
- Add final experimental figures and tables after approved data collection.
- Add uncertainty estimates and repeated-trial statistics.
- Confirm that all safety approvals are documented before experimental work.
- Clean repository paths and ensure reproducible commands are correct.
- Verify all citations and bibliography formatting.
- Compile final PDF and check figures, tables, hyperlinks, and equation numbering.

B Reproducibility Notes

The solver repository is organized so that numerical results can be reproduced from the project root. The current local path is:

```
/Users/a1/Desktop/isef
```

The Python environment used during this draft is Miniforge Python:

```
/Users/a1/miniforge3/bin/python
```

Important commands are:

```
/Users/a1/miniforge3/bin/python -m pytest -q
/Users/a1/miniforge3/bin/python examples/04_taylor_angle_benchmark.py
/Users/a1/miniforge3/bin/python examples/05_shape_optimization_demo.py
/Users/a1/miniforge3/bin/python -m streamlit run app/streamlit_app.py
```

The final repository should include scripts for exporting simulation profiles, processing experimental images, and generating comparison figures.